TEMPLATE (*kindly **duplicate** this file & **replace** this header with your team number & name*)

AI-USE & ETHICS REPORT

Submission Deadline:

Format: PDF

Naming Convention: [Track]_[University]_[TeamName]_AI_Report.pdf

TEAM INFORMATION

Team Name	[insert details] (e.g., DTU 1 - TechGuard ASEAN)
Institution	[insert details] (e.g., Duy Tan University)
Country	[insert details] (e.g., Vietnam)
Track	[ ] Climate Change [ ] Telemedicine [ ] Smart Cities [ ] AI for Education
Project Title	Climate Resilience Copilot: AI-Native Heatwave Electrical-Fire Early Warning for ASEAN Buildings

1. INTRODUCTION

Climate Resilience Copilot addresses a concrete climate-adaptation problem in ASEAN buildings: extreme heatwaves increase cooling demand, overload electrical systems, and may create pre-fire conditions before smoke or flame appears. The project focuses on early warning for dormitories, campuses, public buildings, and dense urban zones exposed to heat-driven infrastructure stress. AI is necessary because the risk is compound: temperature, humidity, active power, and device status must be interpreted together, not as isolated sensor readings. Our objective is to convert distributed IoT telemetry into explainable, auditable, and human-supervised decision support for earlier preventive action.

2. PROBLEM CONTEXT & SOLUTION OVERVIEW

Many ASEAN communities experience rising heat, aging electrical infrastructure, and limited facility-management resources. Building operators often receive fragmented sensor readings or late alarms, while students, staff, and residents may remain exposed until a visible incident occurs. The main stakeholders are facility managers, campus safety staff, building operators, students, and local communities.

The solution integrates Matter and Zigbee telemetry into FIWARE Orion as NGSI-v2 digital-twin entities. The system monitors temperature, humidity, active power, smart-plug status, timestamps, and zone context. An MCP-based resilience agent observes the digital twin, computes warning or critical risk through transparent rules, enriches the result with AI-generated rationale, and recommends response playbooks. When needed, it publishes alerts, requests operator approval, simulates protective actions such as alert signaling or power-isolation, and records ACK/ERROR outcomes. The system is not a certified fire-alarm replacement; it is a climate-resilience decision-support layer for earlier intervention.

3. AI TOOLS & METHODS USED

The prototype uses a hybrid AI architecture. The core safety path is a rule-first risk engine that evaluates heatwave-related compound conditions using explicit thresholds for temperature, active power, and humidity. This deterministic layer provides low latency, traceability, and fail-safe behavior. Above it, an MCP-based AI agent acts as a controlled tool-use layer between AI reasoning and the FIWARE digital-twin environment. Its tools include context query, risk computation, alert publication, operator evaluation, AI explanation, and simulated or approval-gated command invocation.

FIWARE Orion stores the digital-twin state, while FIMAT and the Zigbee bridge normalize telemetry from Matter emulators or optional devices. A configurable LLM endpoint, defined through AI_ENDPOINT, AI_API_KEY, and AI_MODEL, is used only for explanation, operator guidance, scenario interpretation, and response-playbook wording. If the LLM is unavailable, the rule engine still detects risks and publishes alerts.

4. ASSESSMENT OF AI OUTPUT (CRITICAL EVALUATION)

Accuracy. The most safety-critical output is the risk classification: normal, warning, or critical. For this reason, the system does not rely on generative AI for primary detection. Transparent rules compute the baseline risk from actual telemetry, and AI explanations are checked against those rule outcomes. The draft validation plan uses replayable normal, warning, and critical scenarios. Evidence will include timestamped Orion entities, risk-engine logs, AlertEvent records, command ACK/ERROR logs, and dashboard screenshots. Target indicators are detection within five seconds in the demo flow, zero false negatives in predefined critical scenarios, and complete alert/action audit records.

Technical bias. The MVP does not process personal identity data, faces, voice identity, demographic profiles, or sensitive records, so classic demographic bias is minimized. However, technical bias can still occur if thresholds are calibrated only for one room, one sensor brand, or one building type. Heat, humidity, electrical load, and infrastructure quality can vary across ASEAN countries, old dormitories, schools, industrial parks, and public buildings. Therefore, the current thresholds should be treated as controlled-demo parameters, not universal safety standards.

Cultural and regional sensitivity. AI recommendations must fit ASEAN realities: limited budgets, mixed infrastructure ages, different staffing levels, and local emergency procedures. The system avoids overclaiming autonomous control and instead provides cautious operator guidance. Messages should be clear enough for non-technical facility staff.

Linguistic nuance. The MVP may initially use English technical output. For real deployment, alert wording should support Vietnamese and other ASEAN languages. Any AI-generated message must be reviewed to avoid confusing, overly technical, or culturally inappropriate instructions.

5. HUMAN INTERVENTION & JUSTIFICATION

Human intervention is central to the design. Team members define the risk thresholds, scenario labels, response policies, demo scripts, and acceptable command behavior. AI-generated explanations are not accepted automatically; they are compared with the actual telemetry values and deterministic risk results. Sensitive actions, including simulated power isolation or any future hardware bridge, require operator approval and must be recorded with timestamp, reason, risk level, and ACK/ERROR status.

AI alone is insufficient because LLMs can hallucinate, overstate confidence, or ignore local building constraints. Human expertise is also needed to decide whether a recommendation is safe, practical, legally acceptable, and appropriate for a specific campus or facility. The team therefore uses AI for decision support, explanation, and coordination, while preserving human responsibility for real-world safety decisions and emergency escalation.

6. REFLECTION ON AI-HUMAN CO-CREATION

AI helped the team move beyond a passive dashboard. It made raw telemetry easier to interpret, supported faster explanation of compound risks, and helped generate response playbooks for operators. The main risks were hallucinated explanations, loss of local nuance, and over-automation in a safety-sensitive context. The team learned that responsible AI should not replace deterministic safety logic or human judgment. The strongest design is a co-creation model: transparent rules detect critical conditions, AI improves understanding and communication, and humans approve sensitive actions with full auditability and clear operational accountability.

7. CONCLUSION

Climate Resilience Copilot contributes to a more resilient ASEAN by detecting heatwave-related infrastructure stress earlier in buildings where delayed response can increase risk. The solution combines low-cost IoT telemetry, FIWARE digital twins, MCP-based AI orchestration, explainable recommendations, and auditable human-supervised actions. Its ethical value lies not only in using AI, but in limiting AI where necessary: no PII, no opaque autonomous emergency control, and no unsupported claims. Our final position is that AI should strengthen human decision-making, transparency, and preparedness, while communities retain accountability for real-world safety actions and long-term adaptation.

8. APPENDICES (IF NEEDED)

Supporting evidence for the draft submission:

1. Dashboard screenshots showing normal, warning, and critical states.
2. FIWARE Orion entity samples for SensorReading, AlertEvent, and ZoneRisk.
3. MCP tool-use logs showing query context → compute risk → publish alert → request or approve action.
4. AI explanation samples linked to actual telemetry values.
5. Command ACK/ERROR log samples.
6. Demo scenario script for normal, warning, and critical replay.
7. Prompt samples used for AI explanation or operator guidance.
8. Data statement confirming synthetic telemetry, optional Matter/Zigbee device telemetry, and no PII.
9. Ethics checklist confirming human-in-the-loop, no autonomous unsafe control, and audit trail for sensitive actions.

SUBMISSION CHECKLIST FOR TEAMS:

1. **Word Count:** Ensure the body text is between 800-1,000 words.
2. **Ethics First:** Be honest about AI limitations and hallucinations.
3. **Naming Convention:** Ensure the file is named correctly before uploading.
4. **Submission Link:** Team Leaders must upload to the **Final Submission Folder** **FINAL SUBMISSION** **by June 20, 2026.**